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Additional Notes on Finite Element Models of Deformable Featureless Headform

**Clifford C. Chou, Saeed D. Barbat, Nengquan (Nick) Liu,
Gongfu F. Li, Fubang Wu, and Yonglu Zhao**
Ford Motor Co.

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ABSTRACT

Model characteristics of a finite element deformable featureless headform with one to four layers of solid elements for the headform skin are studied using both the LS-DYNA3D and FCRASH codes. The models use a viscoelastic material law whose constitutive parameters are established through comparisons of drop test simulations at various impact velocities with the test data. Results indicate that the one-layer model has a significant distinct characteristic from the other (2-to-4-layer) models, thus requiring different parametric values. Similar observation is also noticed in simulating drop tests with one and two layers of solid elements for the headform skin using PAM-CRASH. When using the same parametric values for the viscoelastic material, both the LS-DYNA3D and FCRASH simulations yield the same results under identical impact conditions and, thereby, exhibit a "functional equivalency" between these two codes.

In addition, the effects of viscoelastic parameters on the headform responses are studied. Further utilization of these models in upper vehicle interior structural simulations yields comparable results regardless of which deformable featureless headform (i.e. 1- to 4-layer models) is used as an impactor.

1.0 INTRODUCTION

The National Highway Traffic Safety Administration (NHTSA) issued a Notice of Proposed Rulemaking [1] and published the final rule on interior head impact requirements as an amendment to FMVSS 201 in August, 1995. NHTSA defined the impact targets, directions and locations on a typical vehicle for compliance verification which the automobile manufacturers may have to test during certification to verify full compliance of the entire vehicle upper interior. Automobile manufacturers are able to obtain design direction and reduce the number of tests in vehicle development by utilizing CAE modeling and simulations. To achieve accuracy in FEA simulation of interior headform impact, development and validation of a featureless headform model is required.

Several deformable headform developments using various non-linear finite element codes have been discussed in other papers [2-8]. A comparison of four models is given in Table 1. Sugita et al. [2] developed a featureless headform using PAM-

CRASH. In their model, the dummy skin was modeled with three layers of solid elements using a crushable foam material law to represent the skin characteristics derived from static compression test data. This model was validated with data from a standard head drop test at 2.68 m/s (6.07 mph). Another rubber-like headform skin model was developed by Bilkhu et al. [3] using LS-DYNA3D. The skin of the headform was modeled with three layers of solid elements and an elastic-plastic hydrodynamic material law based on data derived from a uni-axial quasi-static test. The validation of the model was carried out by comparing the simulated results with those obtained in a 6.71 m/s (15 mph) head drop test.

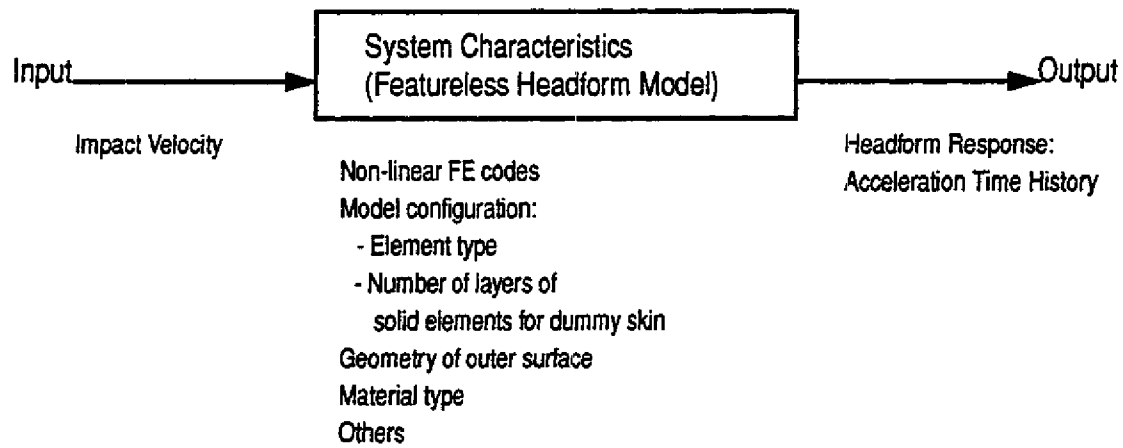
Barbat et al. modeled the dummy skin with one layer and two layers of solid elements using a simple linear viscoelastic material in PAM-CRASH [4] and RADIOSS [5]. Barbat's model as well as those discussed by Chou, Zhao, Huang and Lim using LS-DYNA3D [6,7] and by Liu, Wu, Chou and Zhao using FCRASH [8] were validated against data from headform drop tests at 2.68, 4.02 and 6.71 m/s (6, 9 and 15 mph). In addition, for all those models, the predictions of headform responses in both component and full vehicle level simulations were in excellent agreement with related experiments.

While evaluating the material simulation capabilities in LS-DYNA3D, Chou et al. [6] also developed a featureless headform model using the viscoelastic material. The dummy skin was modeled with solid elements of various layers ranging from one to four. The viscoelastic material parameters of these models were determined based on high velocity head drop tests conducted by Barbat et al., as reported in [4]. These models were translated into FCRASH, Ford internal code, for future applications. The parametric values thus obtained are considered to be the coefficients of "transfer functions" rather than the actual constitutive constants. The effects of number of layers of solid elements on the headform response will be discussed.

A model with a given configuration possesses its own characteristics. This model, when subjected to an impact, produces responses in terms of accelerations time history in our study. The relationship between the input, the model characteristics, and the output can be represented schematically in Figure 1. When the model characteristics are determined based on known input and output, the relationship between the output and the input in terms of these characteristics is called a transfer function.

Table 1: Comparison of Various Deformable Featureless Headform Models (Literature Review)

Developed by	Code Used	Material Type	Skin Model (Solid Element)	Head Drop Test Validation	Reference
Sugita et al.	PAM-CRASH	Crushable Foam	3 Layers	Standard Head Drop Test @ 6 mph	[2] 1995
Bilkhu et al.	LS-DYNA3D	Elastic-Plastic Hydrodynamic	3 Layers	Head Drop Test @ 6 and 15 mph	[3] 1995
Barbat et al.	PAM-CRASH & RADIOSS	Visco-elastic	1 & 2 Layers	Head Drop Test @ 6, 9, 12 and 15 mph	[4,5] 1996
Chou et al.	LS-DYNA3D & FCRASH	Visco-elastic	1-4 Layers	Head Drop Test @ 6, 9, 12 and 15 mph	[6-8] 1995/1996

**Figure 1. Relationship between input, model system characteristics and output.**

The model characteristics are largely influenced by the headform model configuration, interface parameters, the code used in the analysis, etc. This study attempts to determine to what degree these viscoelastic parameters be affected by the headform model configurations, including number of layers of solid elements used and the non-linear finite element code used?

In addition, this paper describes the development and validation of deformable featureless headform models using a viscoelastic material law and its applications to structural component and full vehicle system simulations. Description of the featureless headform models is given in Section 2.0, while the viscoelastic material model is presented in Section 3.0. Head drop tests including the calibration test and higher velocity drop tests are outlined in Section 4.0. Determination of viscoelastic parameters is detailed in Section 5.0. Results of the effects of these parameters on the deformable headform response are discussed in Section 6.0. Section 7.0 presents demonstrations carried out at the sub-system level using the deformable featureless headform. Finally, conclusions are drawn in Sections 8.0.

2.0 DEFORMABLE FEATURELESS HEADFORM

The featureless headform is basically a Hybrid III head without a nose. It consists of a hollow cast aluminum thick shell

which approximates the skull exterior geometry. The occipital portion of the shell is removed and replaced by a back plate for mounting the headform to a propulsion shaft. The aluminum skull is covered with a vinyl rubber skin. The headform assembly is instrumented with a tri-axial accelerometer located at the headform center of gravity to measure its three-dimensional translational accelerations.

A featureless headform is scanned to provide geometry line data for generating the FE model. The deformable featureless headform FE model consists of skull and skin as shown in Figure 2. The aluminum skull, back plate, ballast, and accelerometer fixture are represented by rigid shell elements. The headform skin is modeled with 8-noded solid elements and is represented by a viscoelastic material. In this study, various layers of solid elements, ranging from 1 to 4, are used to model the headform skin as shown in Figure 3. Simulations with these deformable featureless headform models are executed with both LS-DYNA3D and FCRASH codes using similar input data. The material type #14 (Linear Viscoelastic) in FCRASH and material type # 6 (Viscoelastic) in LS-DYNA3D are used and found to be functionally equivalent. The parameters required for the headform skin model are determined based on data obtained from head drop tests at different velocities [4].

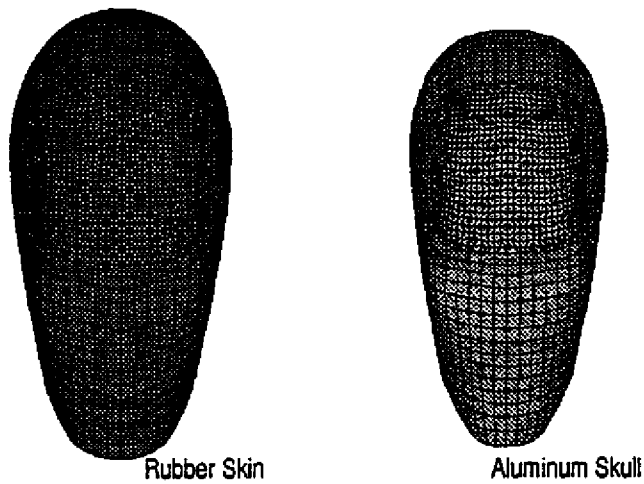


Figure 2. Deformable featureless headform FE model.

3.0 VISCOELASTIC MATERIAL MODEL

Material characterization of the vinyl headform skin is not an easy task. Test results of some recent attempts to achieve this characterization showed that the mechanical response of this material is not only non-linear but also time dependent. Therefore, an appropriate constitutive equation, which takes into account the large deformation and time dependency, can be expressed as, for a uniaxial extension case:

$$\alpha(t) = g(\lambda(t), t) + \int h[\lambda(t), \lambda(s), (t-s)] dt$$

where g and h are the material property functions. Detailed review of rubber viscoelasticity is given by Morman [9]. Use of this model requires extensive material testing. Some tests were conducted to derive property functions based on stress relaxations. Test data so far were non-conclusive, and thus precluded establishment of these functions.

An alternative approach is to use a linear viscoelastic model which takes time-dependency into consideration. This material model, based on shear relaxation behavior, is characterized by the following expression:

$$G(t) = G_{\infty} + (G_0 - G_{\infty})e^{-\beta t}$$

where G_0 = short-time shear modulus, G_{∞} = long-time shear modulus and β = decay constant. In using this material type, the bulk modulus κ is also needed.

This material type is available in both LS-DYNA3D and FCRASH. In our study, G_0 , G_{∞} , β and κ are selected in such a way that the headform model is capable of simulating head drop responses at various velocities up to 6.71 m/s (15 mph).

4.0 HEAD DROP TESTS

HEAD DROP CALIBRATION TESTS:

Prior to impact testing, the featureless headform needs to meet the calibration performance requirement specified by NHTSA. Similarly, the featureless headform FE model needs a "computer" calibration prior to its applications to CAE interior head impact analysis. This physical calibration test will serve as a baseline for component validation of the featureless headform FE model.

In the calibration test, a headform must satisfy the standard Hybrid III head drop test verification in order to be certified for testing. The calibration test involves dropping the headform from a height of 376 mm (14.8 inches) onto a rigid steel smooth surface. The resultant acceleration measured at the C.G. of the headform is to be maintained between 225 G's and 275 G's for acceptance [1]. Results of 30 calibration tests are obtained and their average response as well as the upper and lower envelopes are established as a corridor shown in Figure 4, which will be used as a baseline calibration test for the featureless headform FE model validation.

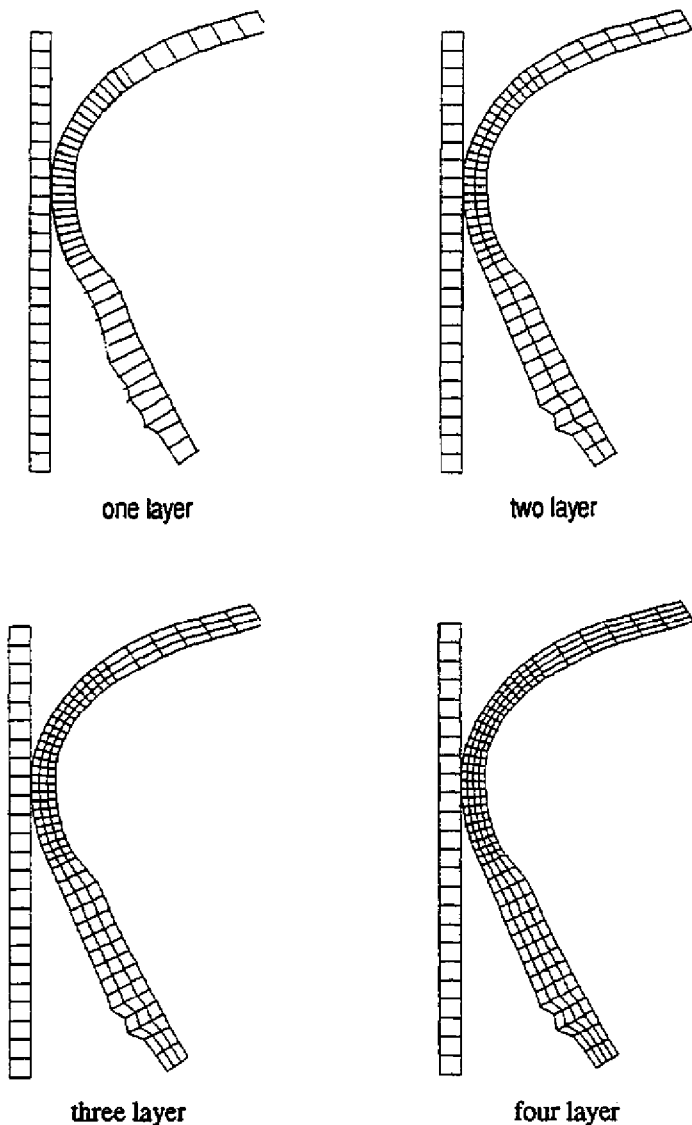


Figure 3. Deformable featureless headform FE models for 1-, 2-, 3- and 4-layer of headform skin.

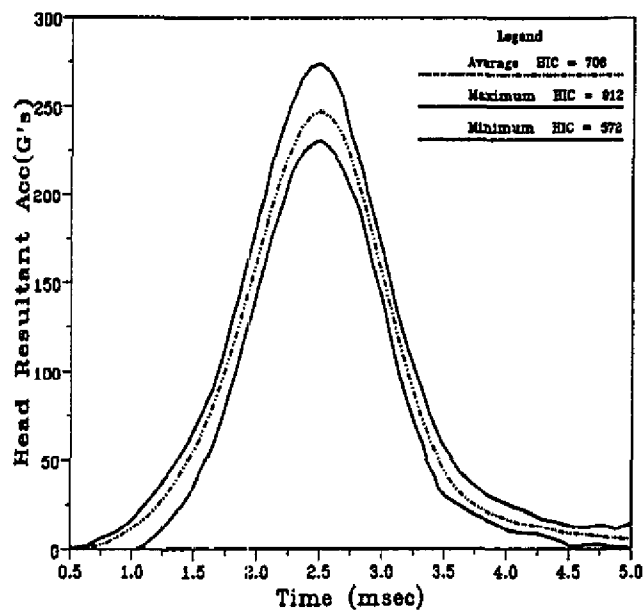


Figure 4. The performance responses of head drop calibration tests.

HIGH VELOCITY HEAD DROP TESTS:

In most simulations using the deformable headform model, the impact velocities are in the proximity of 6.71 m/s (15 mph). It is felt that a headform model validated at 2.68 m/s (6 mph) in a standard drop test may not be adequate for simulating impact at a higher velocity. It is believed that the strain rate effect of the vinyl material is different at the 6.71 m/s (15 mph) actual impact condition than at the 2.68 m/s (6 mph) head drop calibration test. Therefore, a series of higher velocity head drop tests were conducted by Barbat et al. [4] at 4.02 and 6.71 m/s (9, and 15 mph). The translational resultant acceleration for these tests measured at the headform C.G. are shown in Figures 5.

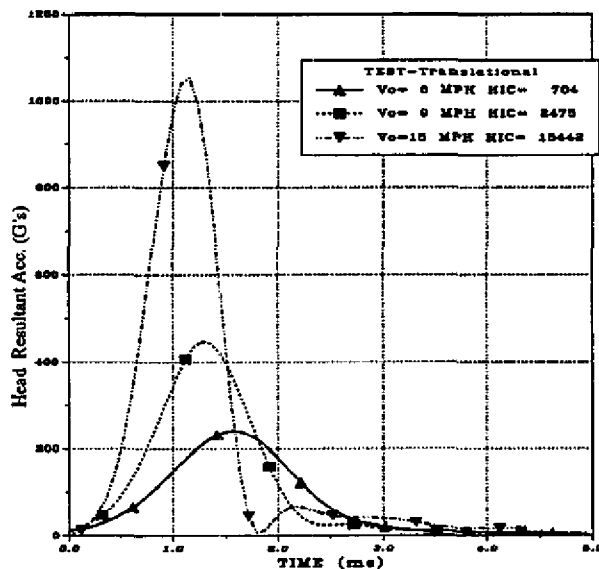


Figure 5. Linear accelerations in head drop tests.

Headform responses obtained from the 2.68, 4.02 and 6.71 m/s (6, 9 and 15 mph) head drop tests are used for selecting the viscoelastic material parameters in the deformable headform model. The parametric values of the viscoelastic model are tuned

in such a way that the simulated responses of the headform dropped at 2.68, 4.02, and 6.71 m/s (6, 9 and 15 mph) are in a good agreement with the test data. Parameters thus determined are considered as coefficients of a transfer function rather than constitutive values as mentioned earlier.

5.0 DETERMINATION OF VISCOELASTIC PARAMETERS

Parameters for the viscoelastic material used in the deformable headform model are determined by comparing the predicted headform responses with data from the head drop tests at various velocities. The process was initially performed by "trial-and-error" procedures and finally by "optimization" techniques through design of experiment method.

PARAMETERS BASED ON HEADFORM DROP CALIBRATION TEST:

To simulate the standard calibration test in the FE analysis, a rigid "stone" wall or a rigid body can be used to represent the rigid plate against which the headform is impacted. The headform is given an initial velocity of 6.3 mph (2.71 m/s), representing the impact velocity from a free fall at a height of 376 mm (14.8 inches) toward the rigid wall. Headform models of varying layers from 1 to 4 are used in the simulations. Various parametric values for G_o , G_{∞} , β and κ associated with different headform models are used. The headform resultant accelerations from these simulations are compared with the calibration test as shown in Figure 6. The simulated responses from the various models lie within the corridor, thus these FEA featureless models satisfy the computer simulated head drop calibration test requirement. Should only the responses of a headform model require to meet the calibration drop test performance, numerous sets of values for these parameters can be found.

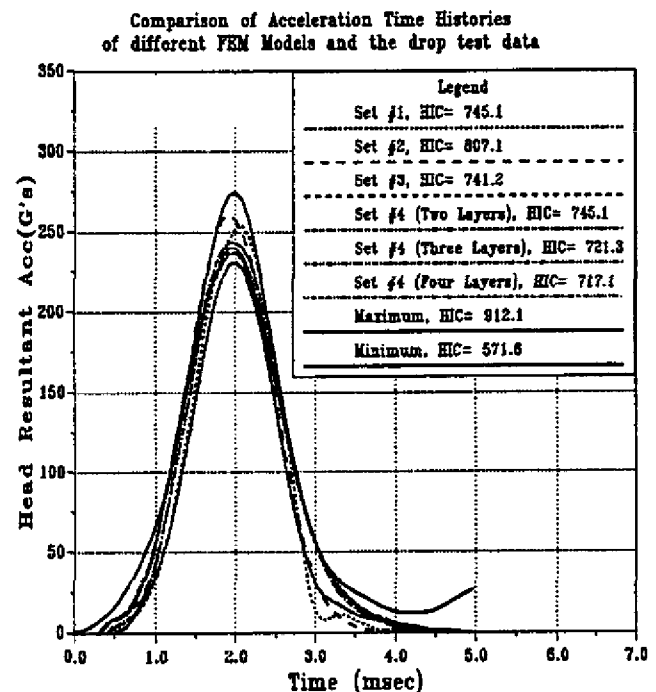


Figure 6. Head resultant accelerations for head drop calibration test.

VALIDATIONS AGAINST HIGHER VELOCITY IMPACT TESTS:

In the head drop test simulation, only a small portion of the solid elements in the forehead are compressed against the rigid surface. When the headform is used to impact foam and/or sheet metal, the headform skin may experience more compression over a larger region due to a higher impact velocity. To determine whether the headform models based on these parametric values will replicate similar responses as observed at higher impact velocities, additional simulations were conducted. These simulations used the same sets of value as those in the calibration tests for all the headform models of 1 to 4 layers at 6.71 m/s (15 mph). Comparisons of headform responses between the simulation and tests at higher velocity are shown in Figure 7. For each data set, the correlations between the tests and the simulated results from their respective models are fairly good. Therefore, the transfer functions are dependent on the "system characteristics" of the model. The system characteristics can be model configuration, formulation of material law and numerical scheme used in codes, etc. However, if a given set of data was applied to various models for simulating headform responses at different velocities, the simulated results reveal general discrepancies with the test data as shown in Figure 8. The discrepancies are particularly noticeable in the simulated results produced by the one-layer headform model. In addition, simulations obtained from two-, three- and four-layer models also yield varying degrees of correlation, although the discrepancies decrease as the layers of solid elements used in the model increase.

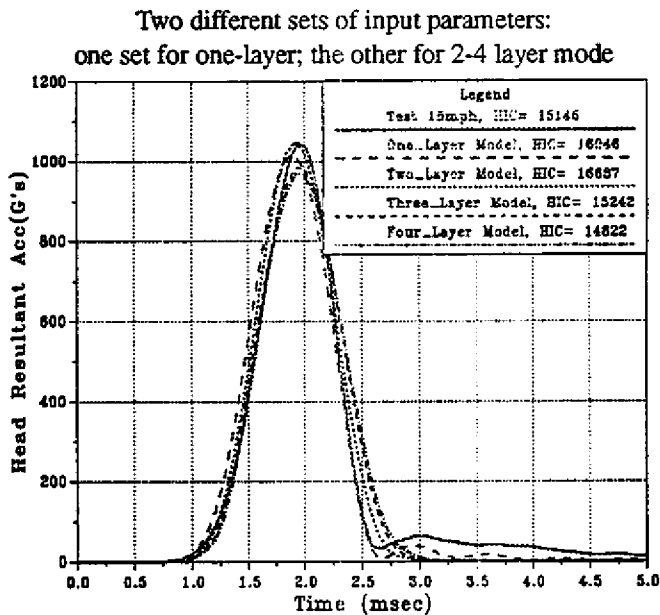
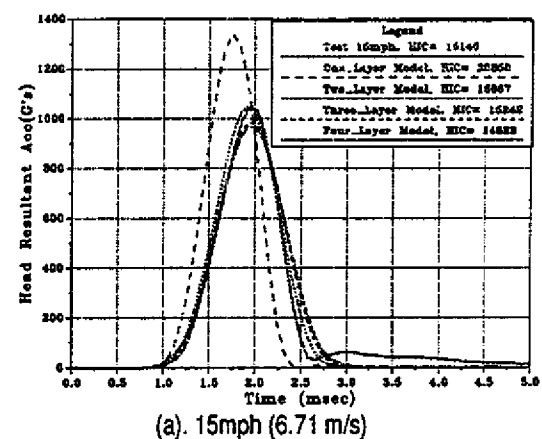
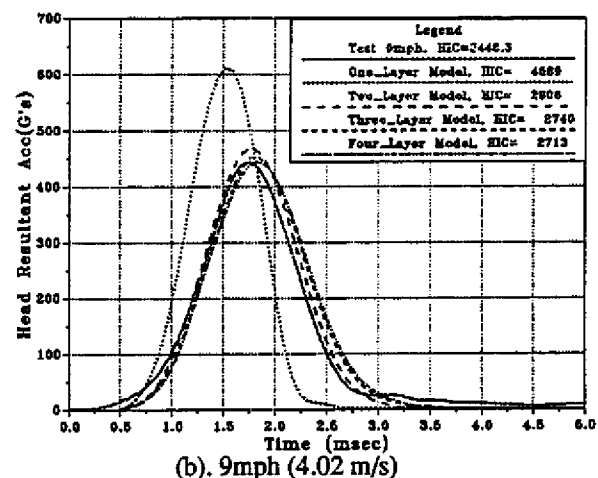


Figure 7. Head resultant accelerations for high velocity drop test.

Same sets of input parameters for 1-4 layer models



Same sets of input parameters for 1-4 layer models



Same sets of input parameters for 1-4 layer models

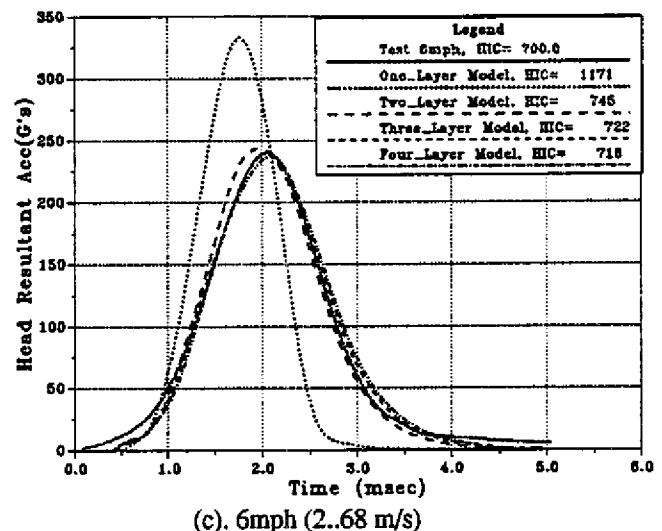


Figure 8. Head resultant accelerations for drop tests at 6.71, 4.02 and 2.68 m/s.

An examination of results shown in Figure 9 reveals that the 2-4 layers deformable headforms give accurate predictive trends at 2.68, 4.02 and 6.71 m/s (6, 9 and 15 mph) test conditions. It is also noted that when the parametric values used in the 2-4 layers models are applied to the one-layer model, stiffer responses are observed.

A very similar behavior was also observed during the development of the featureless headform model using PAM-CRASH code [4]. Two models, with 1 and 2 layers of the solid elements of the headform skin, were developed with two different sets of the viscoelastic parameters. Figure 10 show a very good correlation of the headform responses between the model predictions (2 layers headform skin model) and those obtained from head drop tests at 2.68, 4.02 and 6.71 m/s (6, 9 and 15 mph). Stiffer responses are also observed when applying the viscoelastic parameters used for the 2 layers headform skin model to the one layer headform skin model in simulating high speed drop tests as shown in Figures 11a - 11c.

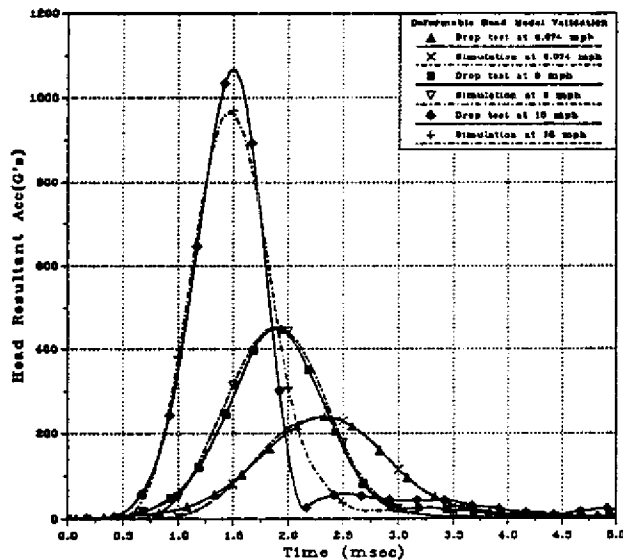
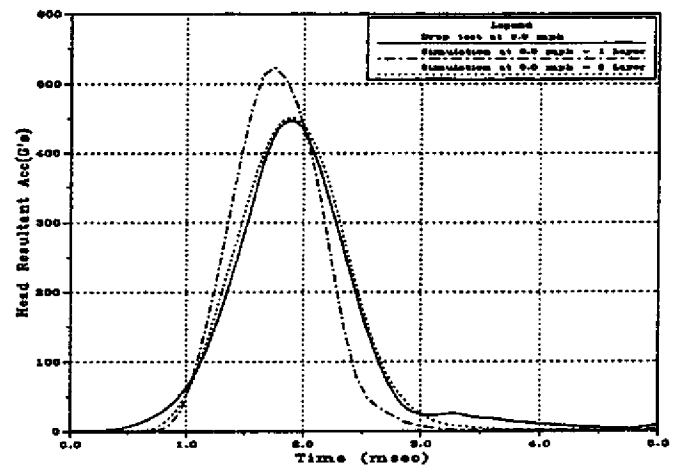
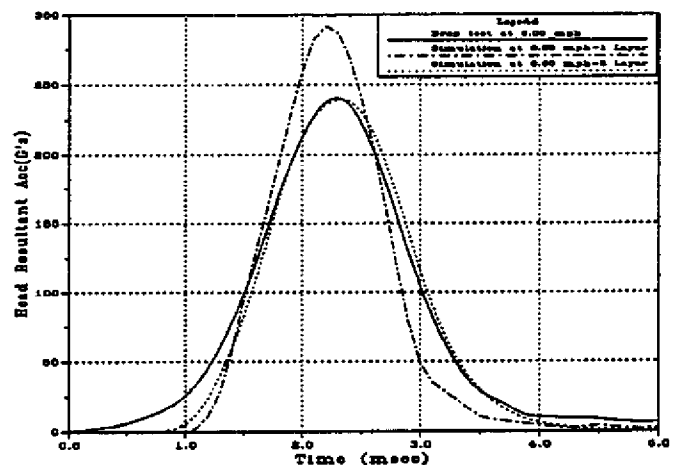


Figure 9. Head resultant accelerations for high velocity drop tests using the 2-layers headform model and PAM-CRASH code.

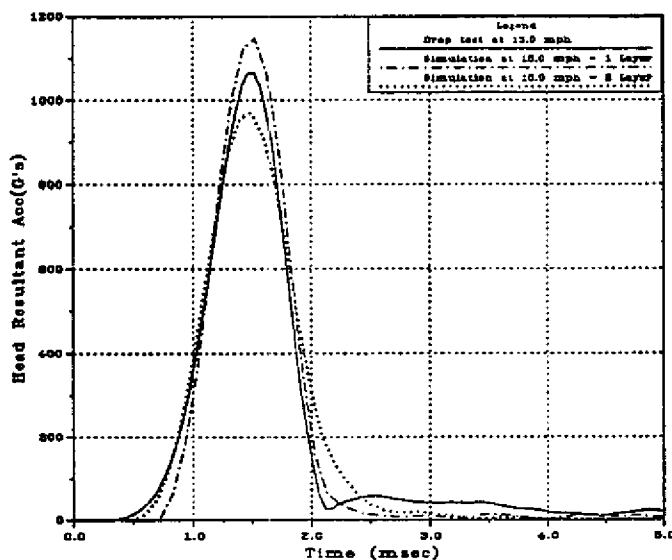


(b) 4.02 m/s (9 mph)



(c) 2.68 m/s (6 mph)

Figure 10. Head resultant acceleration at 2.68, 4.02 and 6.71 m/s head drop using PAM-CRASH



(a) 6.71 m/s (15 mph)

6.0 RESULTS AND DISCUSSIONS

LS-DYNA3D vs. FCRASH:

The LS-DYNA3D and FCRASH models have the same geometry, element type, connectivity, and use similar parameters as closely as possible. Results from simulations of the headform drop tests at 2.68, 4.02 and 6.71 m/s (6, 9 and 15 mph) are compared with the test data as shown in Figures 11-14. Comparisons indicate that the simulation results between the LS-DYNA3D and the FCRASH are about the same. Therefore, it can be concluded that the two codes are functional equivalent in modeling the deformable headform in these test conditions.

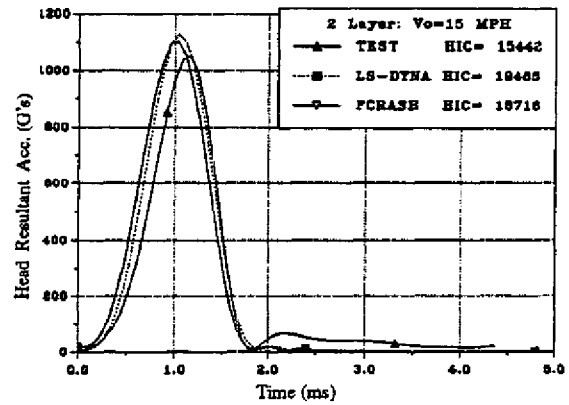
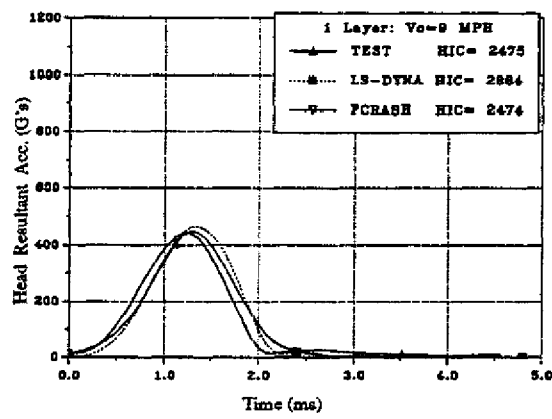
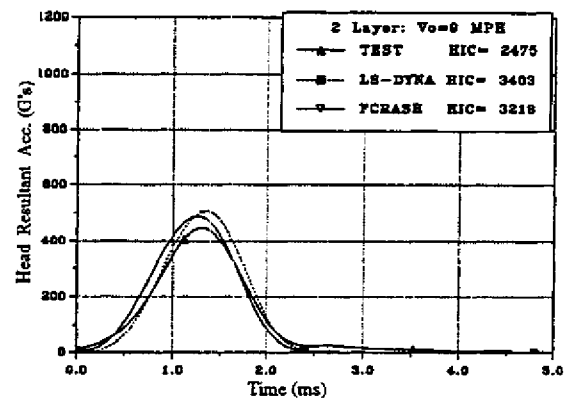
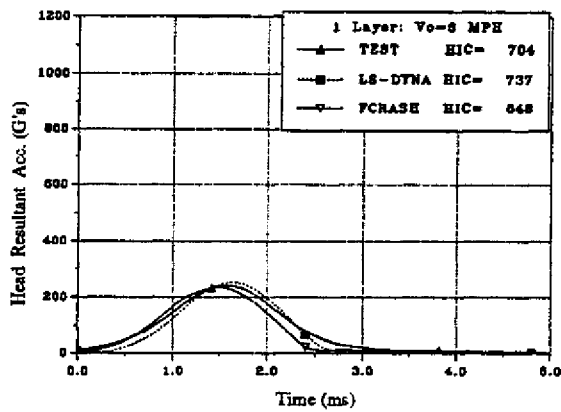


Figure 12. Comparison of LS-DYNA3D and FCRASH in 2-layer models.

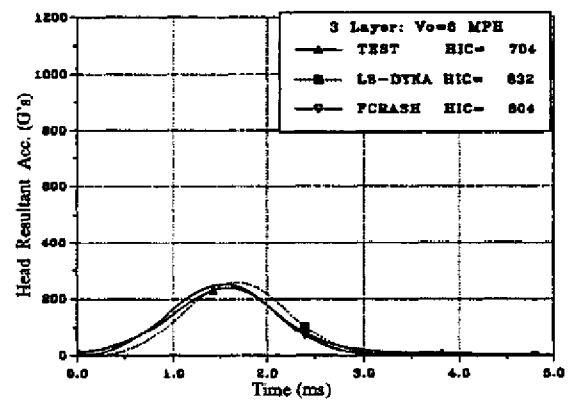
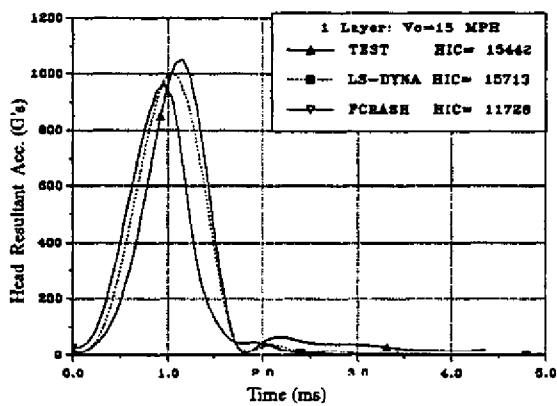
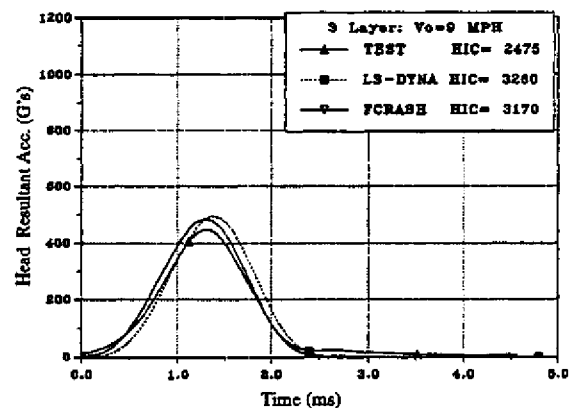
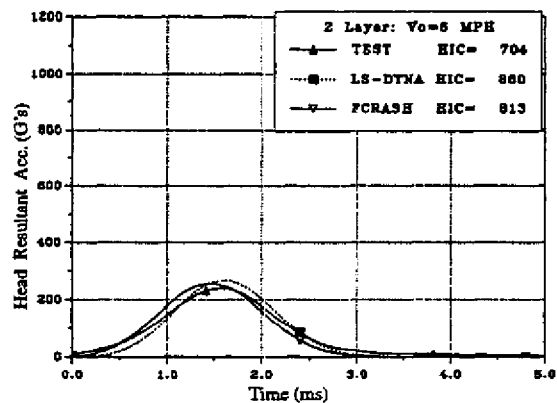


Figure 11. Comparison of LS-DYNA3D and FCRASH in 1-layer models.



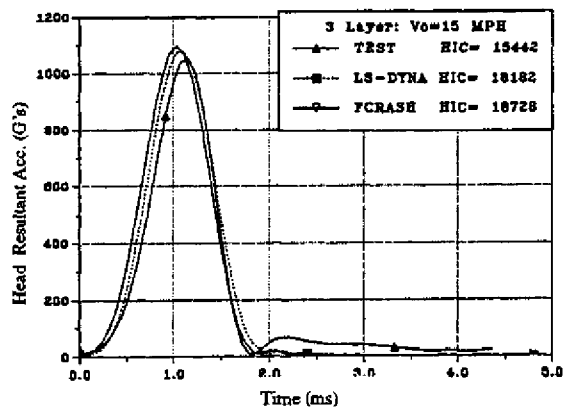


Figure 13. Comparison of LS-DYNA3D and FCRASH in 3-layer models.

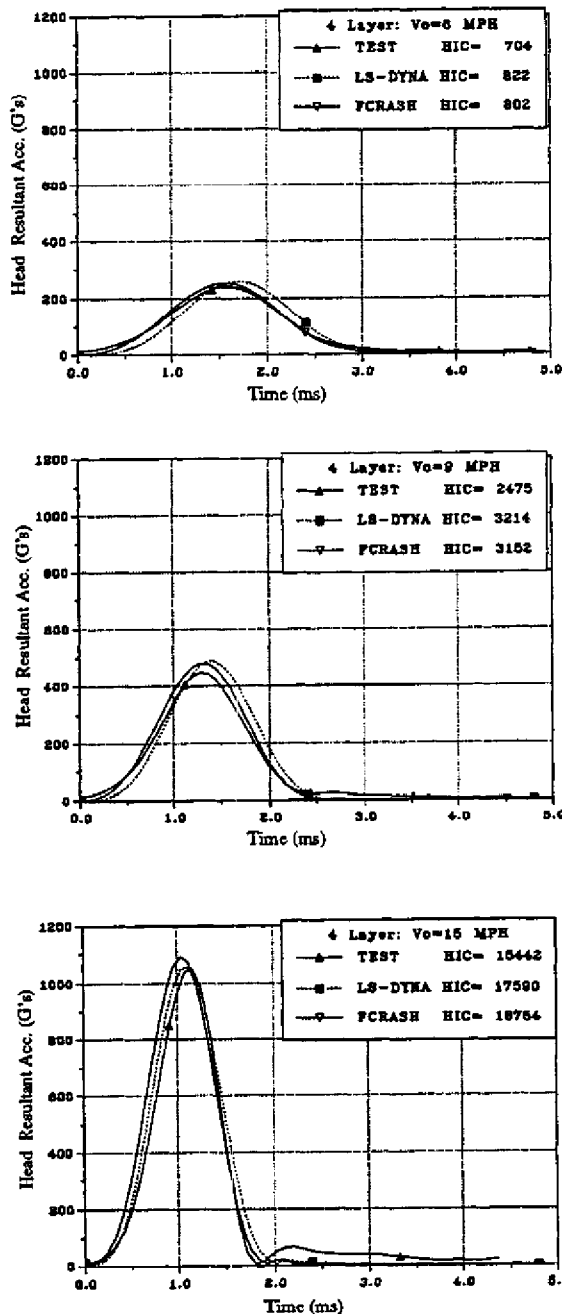


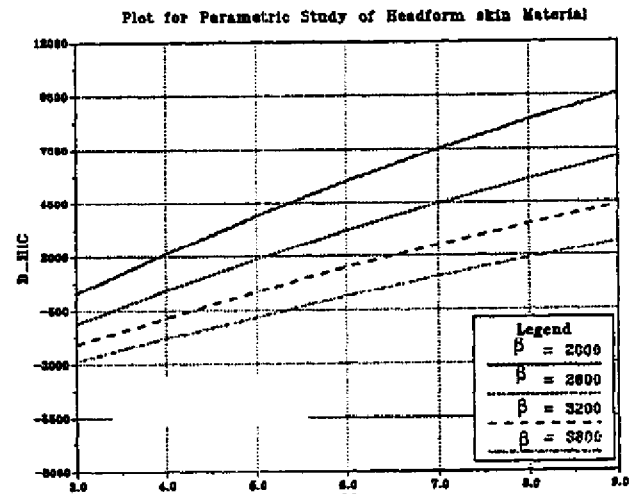
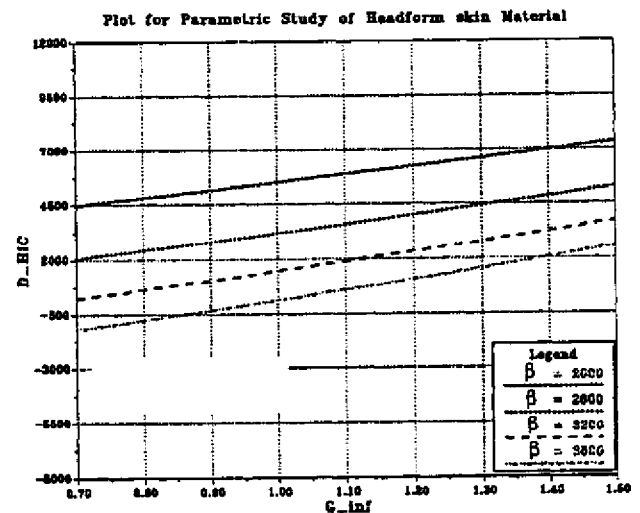
Figure 14. Comparison of LS-DYNA3D and FCRASH in 4-layer models.

EFFECTS OF PARAMETERS G_0 , G_∞ , β , κ :

G_0 , G_∞ , β and κ are four parameters that play a key role in the headform response. The effects of each parameter on the headform responses are studied based on the differences in HIC calculation between the simulated results and test data at 6.71 m/s (15 mph) only, denoted here by D_HIC , and are discussed below:

(i) The Short and Long Term Moduli G_0 , G_∞ :

Given a κ (bulk modulus), D_HIC based on head drop simulations and tests can be expressed as a function of G_0 for a given G_∞ in terms of families of β curves as shown in Figure 15. Results show that for a given β , D_HIC increases as G_0 increases, and so does the effect of G_∞ on D_HIC , as shown in Figure 16. These two parameters dominate the peak value in the headform response.

Figure 15. Effect from parameters G_0 and β , while G_∞ is given.Figure 16. Effect from parameters G_∞ and β , while G_0 is given.

(ii) Effect of Parameter β :

Referring to Figures 15 and 16, results indicate that, for a given G_0 , D_{HIC} varies inversely proportional to β , i.e. D_{HIC} decreases as β increases. In addition, an increase in β would reduce the duration of the response, thus improving the fitness of the simulation shape to the test curve. It is noticed however, that the reduction of the pulse duration is not due to the effect of β only. This can be observed in Figure 17 that the reduction of the acceleration pulse duration is achieved with an increased β value, by adjusting the parameters G_0 and G_∞ in order to keep the same peak accelerations.

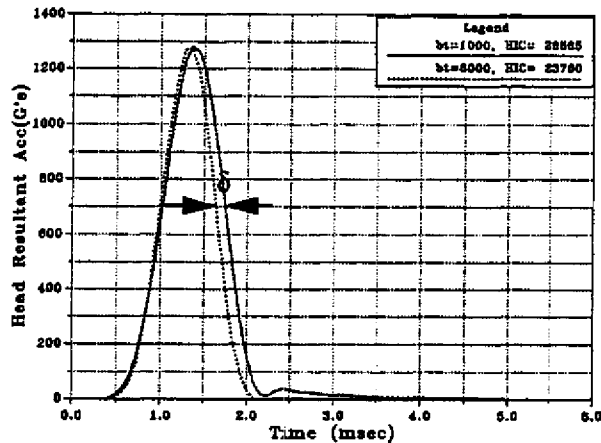


Figure 17. Acceleration duration reduced as β increased, while keeping the same peak acceleration.

(iii) Effect of Parameter κ :

The effect of parameter κ on D_{HIC} is plotted in Figure 18. It can be seen that the parameter κ does not affect the response as significantly as other parameters.

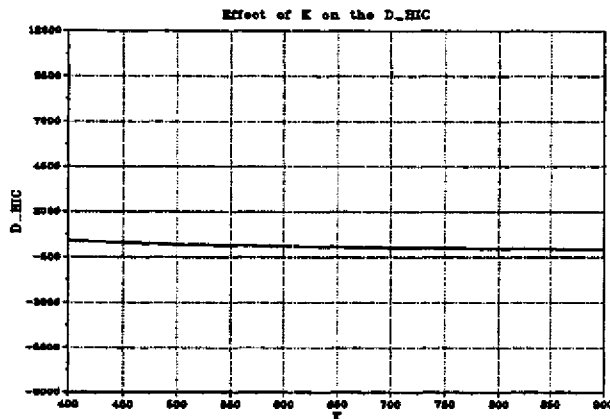


Figure 18. Effect of K on the D_{HIC} .

EFFECT OF LAYERS IN THE FEA MODEL

As already discussed in Section 5.0, the number of layers in the FEA model affects the headform acceleration response. The one-layer model has significant discrepancy in acceleration pulse compared to those of the 2-4 layers models if the parametric values for the 2-4 layers models are used. This effect

is plotted in Figure 19, based on results from a 6.71 m/s (15 mph) head drop simulation. Using D_{HIC} as an error measure, results shown in Figure 19 indicate that the one layer model over predicts the HIC value significantly when compared to those of the 2-4 layer models. The 2-layers model shows slightly over-predicting and the four-layer model under-predicting of the responses. The above trend analysis is based on the same input of viscoelastic parameter. By excluding the one-layer model, overall analysis indicates that a 2-layers model will give a conservative result over other models (i.e. 3 or 4-layers models).

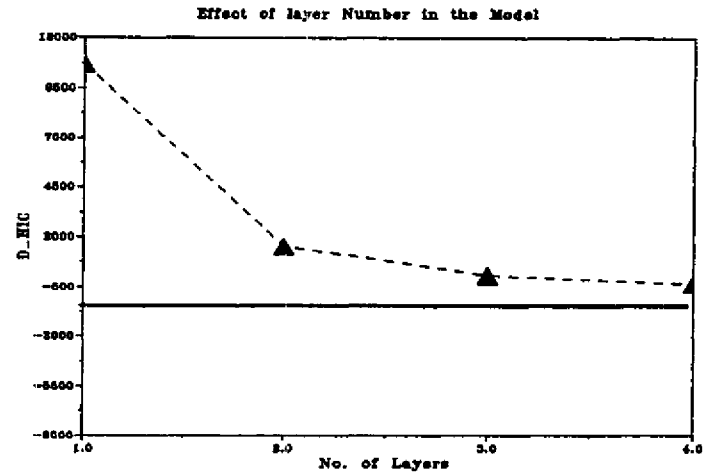


Figure 19. Effect of layers in the FEA model.

CPU TIME:

Computation time was also studied for various headform models with different numbers of layers. Under a simulated head drop test condition, the CPU time required on HP 700 series workstations are plotted in Figure 20. It is seen that the CPU time increases as the numbers of layers of solid elements used for modeling the headform skin increase. To provide an affordable CPU and a desired accuracy in predicting headform responses for head impact analysis, headform models with two layers of solid elements are deemed to be adequate.

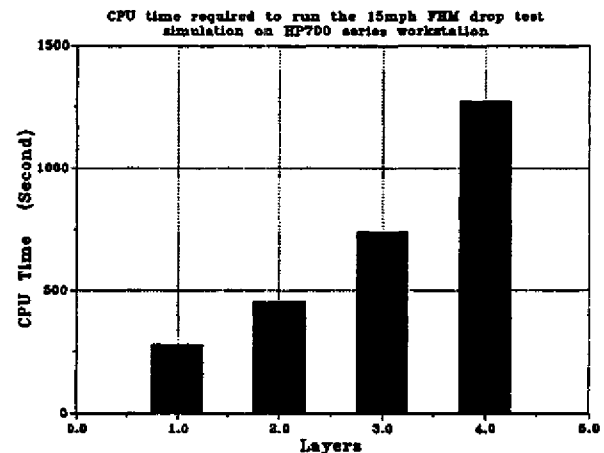


Figure 20. CPU time for simulating the drop test using models with different layers.

BRIEF SUMMARY OF HEADFORM MODEL DEVELOPMENT:

Various deformable headform models were developed using LS-DYNA3D, PAM-CRASH and FCRASH codes. These models were validated against high velocity head drop test data, and were found to be capable of simulating accurately the kinematics of the headform in these test conditions. Use of these models is further demonstrated in the structural sub-system simulations to be discussed next.

7.0 APPLICATIONS OF HEADFORM MODEL

The validated featureless headform models (1 to 4 layers skin) with their respective parameters are used to simulate responses upon impacting sub-system models consisting of sheet metal (S/M) pillar with plastic trim panel and foam padding.

The trim and foam modeling methodology [4 10-11] includes dynamic material characterization and dynamic response correlation. Preliminary polypropylene (PP) and ABS trim models, DYLLITE, DYTHERM, ARPRO and POLYURETHANE foam models are available. The trim is 2.5 mm thick and modeled with shell elements. The property of PP is characterized by tensile testing quasi-statically (12.7 mm/min or 0.5 inches/min) and dynamically at 4.47 m/s (10 mph) and 6.71 m/s (15 mph). The dynamic properties are used in the model via material type #24 in LS-DYNA3D and #2 in FCRASH. Node connectivity is used to tie trim nodes with those of the sheet metal along the flange.

This FE model of the integrated trim/foam/sheet metal, as shown in Figure 21, will be used for trend analysis here. No experimental data are available for correlation. The deformable featureless headform (one-layer to four-layers) is used and the impact velocity is 6.71 m/s (15 mph). Comparisons of the predicted responses between LS-DYNA3D and FCRASH for headform impacts with a sheet metal structure and a foam block are shown in Figure 22. It is noted again that the FCRASH result is similar to that of LS-DYNA3D.

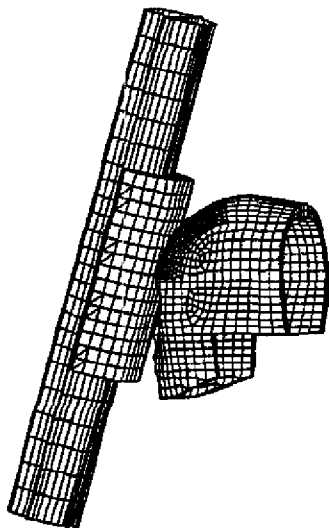
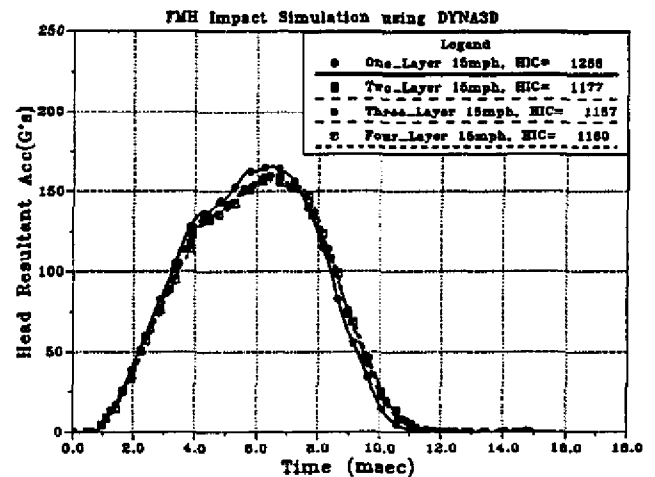
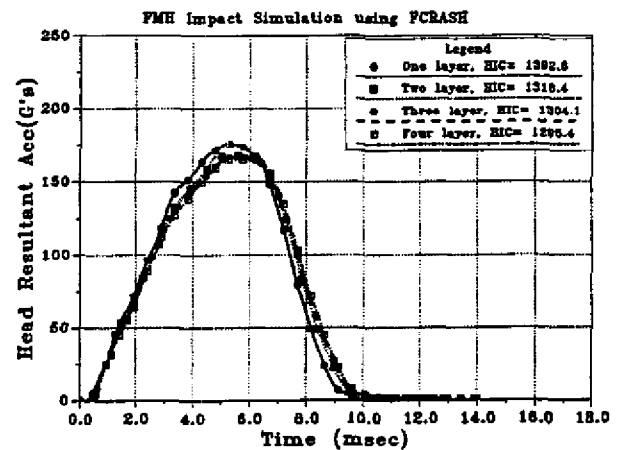


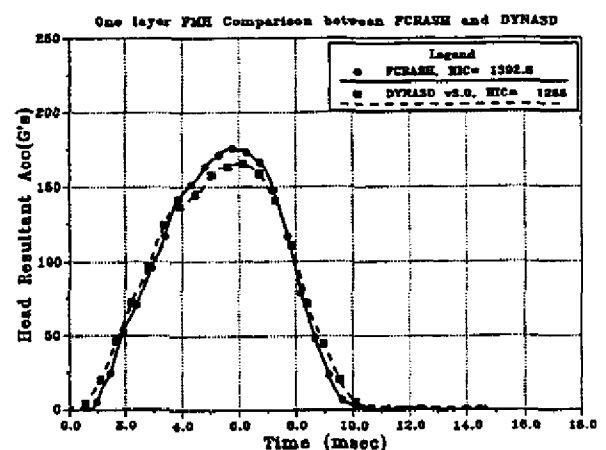
Figure 21. The integrated trim/foam/sheet metal model.



(a) One to four layer headform - LS-DYNA3D



(b) One to four layer headform - FCRASH



(c) One layer headform - LS-DYNA3D and FCRASH

8.0 CONCLUSIONS

Based on viscoelastic material characteristics, deformable featureless headform (DFH) models with various layers of solid elements for the dummy head skin were developed in LS-DYNA3D, PAM-CRASH and FCRASH codes. The number of layers of solid elements used in the model ranged from one to four layers. In the development process, parameters of these featureless headform models were established through comparisons of drop test simulations at various impact velocities with the test data. It was found that two sets of parameters were needed: one for the one-layer model and the other for the 2-to-4-layers models.

It is remarked here that the parametric values used in this study are considered as coefficients of "transfer function", which depends on "model characteristics". Apparently, the one-layer model has a significantly distinct characteristic from the other (2-to-4-layers) models, thus yielding a stiffer response.

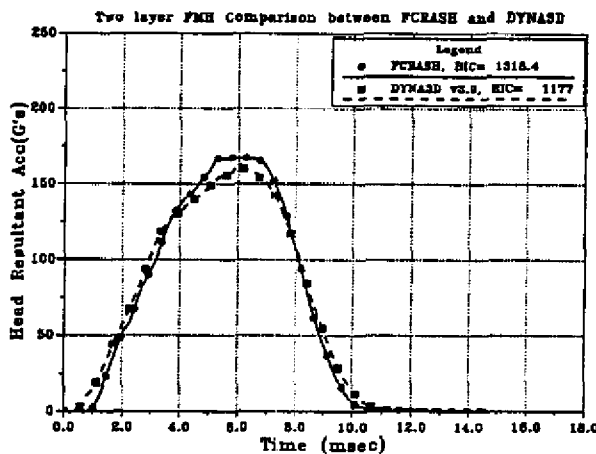
Other findings regarding the DFH models are:

- Using same parametric values for the viscoelastic material, both LS-DYNA3D and FCRASH yielded the similar simulation results under identical impact conditions. However, one should not expect that the parameters established from LS-DYNA3D/FCRASH model will work equally well in models developed using other non-linear codes.
- Both the short and long term moduli, G_0 and G_∞ , affect the peak value of the headform acceleration pulses, while β term influences the duration of the acceleration pulses. The larger the β value, the shorter the acceleration duration, if the peak acceleration is to be maintained.
- Computation time in terms of CPU increases with the number of layers of solid elements used in the model.

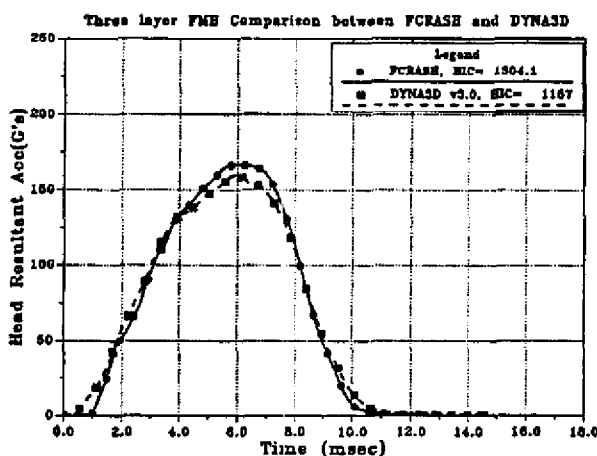
In addition, the DFH models were also used in featureless headform impacts with sheet metal, foam and trim. Simulation results were comparable regardless of which DFH model (i.e. 1- to 4-layer models) was used as an impactor. Currently, the 2 layers model is extensively used in vehicle applications for interior head impact analyses.

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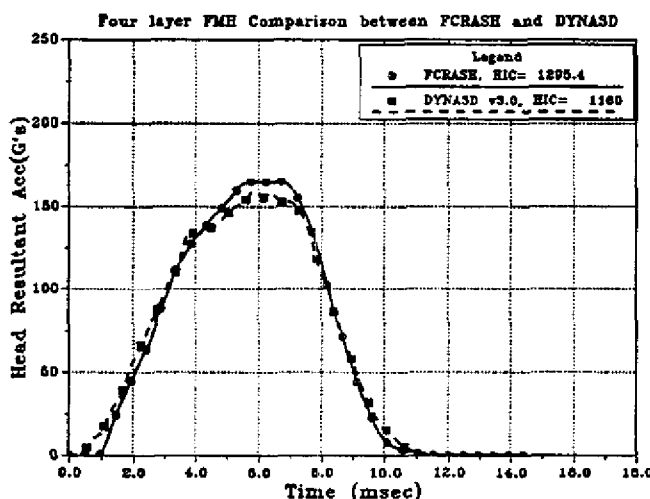
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(d) Two layer headform - LS-DYNA3D and FCRASH



(e) Three layer headform - LS-DYNA3D and FCRASH



(f) Four layer headform - LS-DYNA3D and FCRASH

Figure 22. Simulation results - the integrated trim/foam/sheet metal component validations.

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